

1. General Description

The EM74AUP1G07 is a single buffer with open-drain output. Schmitt-trigger action at all inputs makes the circuit tolerant of slower input rise and fall times. This device ensures very low static and dynamic power consumption across the entire V_{CC} range from 0.8 V to 3.6 V. This device is fully specified for partial power down applications using I_{OFF} . The I_{OFF} circuitry disables the output, preventing the potentially damaging backflow current through the device when it is powered down.

2. Features and Benefits

- Wide supply voltage range from 0.8 V to 3.6 V
- High noise immunity
- CMOS low power dissipation
- Low static power consumption; $I_{CC} = 0.9 \mu A$ (maximum)
- Low noise overshoot and undershoot < 10 % of V_{CC}
- Overvoltage tolerant inputs to 3.6 V
- I_{OFF} circuitry provides partial Power-down mode operation
- Latch-up performance exceeds 200 mA
- Complies with JEDEC standards:
 - JESD8-12 (0.8 V to 1.3 V)
 - JESD8-11 (0.9 V to 1.65 V)
 - JESD8-7 (1.65 V to 1.95 V)
 - JESD8-5 (2.3 V to 2.7 V)
 - JESD8C (2.7 V to 3.6 V)
- ESD protection:
 - HBM: ANSI/ESDA/JEDEC JS-001 Class 2 exceeds 3000 V
 - CDM: ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options
- Specified from -40 °C to +85 °C and -40 °C to +125 °C

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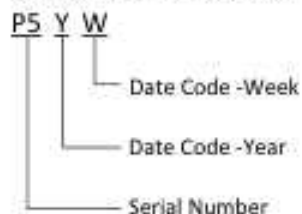
3. Ordering Information

Table 1. Ordering information

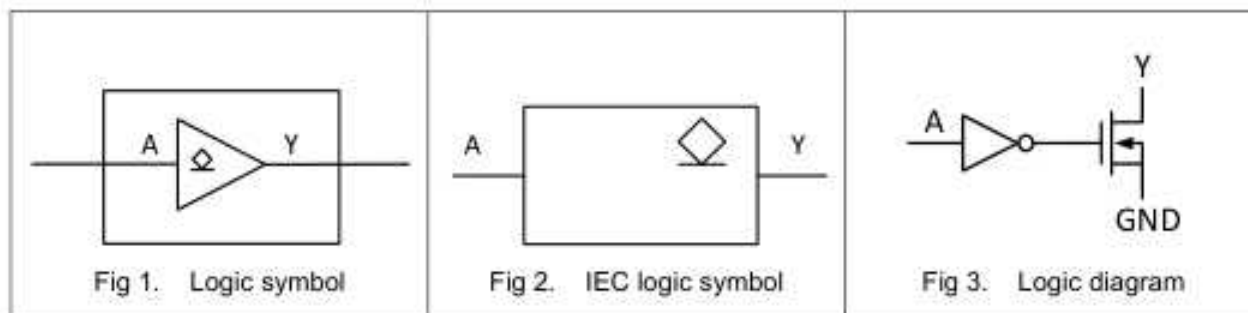
Type number	Topside marking	Package		
		Name	Description	Quantity
EM74AUP1G07GV	P5YW	SOT23-5L	SOT23 package, 5 pins 2.92 mm × 1.6 mm; 1.25 mm (Max) height	3000
EM74AUP1G07GW	P5YW	SOT353	SOT353 package, 5 pins 2.1 mm × 1.25 mm; 1.1 mm (Max) height	3000
EM74AUP1G07GS	P5	DFN1x1-6L	DFN1×1 package, 6 pins 1 mm × 1 mm; 0.42 mm (Max) height	3000
EM74AUP1G07GM	P5YW	DFN1x1.45-6L	DFN1.45×1 package, 6 pins 1.45 mm × 1 mm; 0.6 mm (Max) height	3000
EM74AUP1G07GX	P5	DFN0.8x0.8-5L	DFN0.8×0.8 package, 5 pins 0.8 mm × 0.8 mm; 0.4 mm (Max) height	3000

MARKING INFORMATION

NOTE: YW = Date Code.

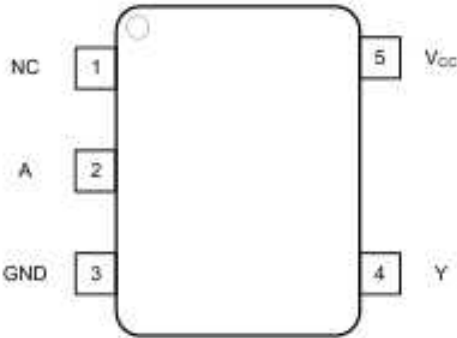
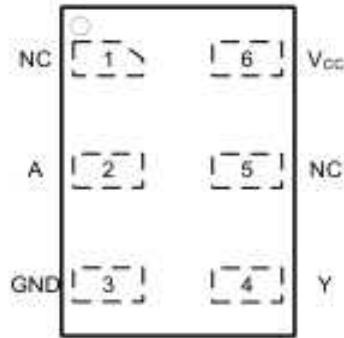
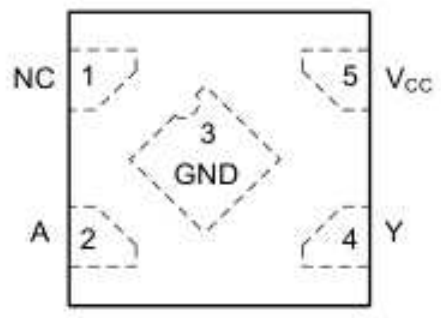


4. Function Diagram



5. Pinning Information

5.1 Pin map

 Fig 4. Top view pin configuration SOT23-5 and SOT353	 Fig 5. Top view pin configuration DFN6L
 Fig 6. Top view pin configuration DFN5L	

5.2 Pin description

Table 2. Pin description

Symbol	Pin		Description
	SOT23-5, SOT353, DFN5L	DFN6L	
NC	1	1	not connected
A	2	2	data input
GND	3	3	ground(0V)
Y	4	4	data output
NC	-	5	not connected
V _{cc}	5	6	supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; Z = high-impedance OFF state.

Input	Output
A	Y
L	L
H	Z

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	4.6	V
I_{IK}	input clamping current	$V_I < 0$ V	-50		mA
V_I	input voltage	[1]	-0.5	4.6	V
I_{OK}	output clamping current	$V_O < 0$ V	-50		mA
V_O	output voltage	Active mode and Power-down mode [1]	-0.5	4.6	V
I_O	output current	$V_O = 0$ V to V_{CC}		20	mA
I_{CC}	supply current			50	mA
I_{GND}	ground current		-50		mA
P_{tot}	total power dissipation	$T_{amb} = -40$ °C to $+125$ °C		250	mW
T_{stg}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{CC}	supply voltage		0.8		3.6	V
V_I	input voltage		0		3.6	V
V_O	output voltage	Active mode	0		3.6	V
		Power-down mode; $V_{CC} = 0$ V	0		3.6	V
T_{amb}	ambient temperature		-40	25	125	°C
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 0.8$ V to 3.6 V	0		200	ns/V

9. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{IH}	HIGH-level input voltage	$V_{CC} = 0.8 \text{ V}$	$0.7V_{CC}$			$0.75V_{CC}$		V
		$V_{CC} = 0.9 \text{ V to } 1.95 \text{ V}$	$0.65V_{CC}$			$0.7V_{CC}$		V
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.6			1.6		V
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	2.0			2.0		V
V_{IL}	LOW-level input voltage	$V_{CC} = 0.8 \text{ V}$			$0.3V_{CC}$		$0.25V_{CC}$	V
		$V_{CC} = 0.9 \text{ V to } 1.95 \text{ V}$			$0.35V_{CC}$		$0.3V_{CC}$	V
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$			0.7		0.7	V
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$			0.9		0.9	V
V_{OL}	LOW-level output voltage	$V_I = V_{IH} \text{ or } V_{IL}$						
		$I_O = 20\mu\text{A}; V_{CC} = 0.8 \text{ V to } 3.6 \text{ V}$			0.10		0.11	V
		$I_O = 1.1 \text{ mA}; V_{CC} = 1.1 \text{ V}$			$0.3V_{CC}$		$0.33V_{CC}$	V
		$I_O = 1.7 \text{ mA}; V_{CC} = 1.4 \text{ V}$			0.37		0.41	V
		$I_O = 1.9 \text{ mA}; V_{CC} = 1.65 \text{ V}$			0.35		0.39	V
		$I_O = 2.3 \text{ mA}; V_{CC} = 2.3 \text{ V}$			0.33		0.36	V
		$I_O = 3.1 \text{ mA}; V_{CC} = 2.3 \text{ V}$			0.45		0.50	V
		$I_O = 2.7 \text{ mA}; V_{CC} = 3.0 \text{ V}$			0.33		0.36	V
		$I_O = 4.0 \text{ mA}; V_{CC} = 3.0 \text{ V}$			0.45		0.50	V
I_i	Input leakage current	$V_I = \text{GND to } 3.6 \text{ V}; V_{CC} = 0 \text{ V to } 3.6 \text{ V}$			± 0.5		± 0.75	μA
I_{O2}	OFF-state output current	$V_I = V_{IH} \text{ or } V_{IL}; V_O = 0 \text{ V to } 3.6 \text{ V}; V_{CC} = 3.6 \text{ V}$			± 0.5		± 0.75	
I_{OFF}	power-off leakage current	$V_{CC} = 0 \text{ V}; V_I \text{ or } V_O = 0 \text{ V to } 3.6 \text{ V}$			± 0.5		± 0.75	μA
ΔI_{OFF}	additional power-off leakage current	$V_{CC} = 0 \text{ V to } 0.2 \text{ V}; V_I \text{ or } V_O = 0 \text{ V to } 3.6 \text{ V}$			± 0.6		± 0.75	μA
I_{CC}	supply current	$V_I = V_{CC} \text{ or } \text{GND}; I_O = 0 \text{ A}; V_{CC} = 0.8 \text{ V to } 3.6 \text{ V}$			0.9		1.4	μA

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ΔI_{CC}	additional supply current	per pin ; $V_{CC} = 3.3\text{ V}$; $V_I = V_{CC} - 0.6\text{ V}$; $I_O = 0\text{ A}$			50		75	μA
C_i	input capacitance	$V_{CC} = 0\text{ V to } 3.6\text{ V}$; $V_I = \text{GND or } V_{CC}$		2.0				pF
C_o	output capacitance	$V_O = \text{GND}$; $V_{CC} = 0\text{ V}$		3.0				pF

[1] All typical values are measured at $T_{amb} = 25^\circ\text{C}$.

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 8.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
C _L = 5 pF								
t _{pd}	propagation delay	A to Y; see Fig. 7 [2]						
		V _{CC} = 0.8 V		78.4				ns
		V _{CC} = 1.1 V to 1.3 V		10.7				ns
		V _{CC} = 1.4 V to 1.6 V		7.8				ns
		V _{CC} = 1.65 V to 1.95 V		5.7				ns
		V _{CC} = 2.3 V to 2.7 V		3.8				ns
		V _{CC} = 3.0 V to 3.6 V		3.2				ns
C _L = 10 pF								
t _{pd}	propagation delay	A to Y; see Fig. 7 [2]						
		V _{CC} = 0.8 V		80.1				ns
		V _{CC} = 1.1 V to 1.3 V		12.5				ns
		V _{CC} = 1.4 V to 1.6 V		8.5				ns
		V _{CC} = 1.65 V to 1.95 V		6.2				ns
		V _{CC} = 2.3 V to 2.7 V		4.1				ns
		V _{CC} = 3.0 V to 3.6 V		3.5				ns
C _L = 15 pF								
t _{pd}	propagation delay	A to Y; see Fig. 7 [2]						
		V _{CC} = 0.8 V		82.3				ns
		V _{CC} = 1.1 V to 1.3 V		13.7				ns
		V _{CC} = 1.4 V to 1.6 V		9.0				ns
		V _{CC} = 1.65 V to 1.95 V		6.7				ns
		V _{CC} = 2.3 V to 2.7 V		4.5				ns
		V _{CC} = 3.0 V to 3.6 V		3.6				ns
C _L = 30 pF								
t _{pd}	propagation delay	A to Y; see Fig. 7 [2]						
		V _{CC} = 0.8 V		97.2				ns
		V _{CC} = 1.1 V to 1.3 V		17.3				ns

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		$V_{CC} = 1.4 \text{ V to } 1.6 \text{ V}$		10.8				ns
		$V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$		8.4				ns
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$		5.2				ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$		4.5				ns
$C_L = 5 \text{ pF, } 10 \text{ pF, } 15 \text{ pF and } 30 \text{ pF}$								
C_{PD}	power dissipation capacitance	$f_i = 1 \text{ MHz};$ $V_i = \text{GND to } V_{CC}$ [3]						
		$V_{CC} = 0.8 \text{ V}$		0.5				pF
		$V_{CC} = 1.1 \text{ V to } 1.3 \text{ V}$		0.5				pF
		$V_{CC} = 1.4 \text{ V to } 1.6 \text{ V}$		0.6				pF
		$V_{CC} = 1.65 \text{ V to } 1.95 \text{ V}$		0.6				pF
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$		0.7				pF
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$		0.9				pF

[1] All typical values are measured at nominal V_{CC} and $T_{amb} = 25^\circ\text{C}$.

[2] t_{PHL} is the same as t_{R1} and t_{R2} .

[3] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \sum(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

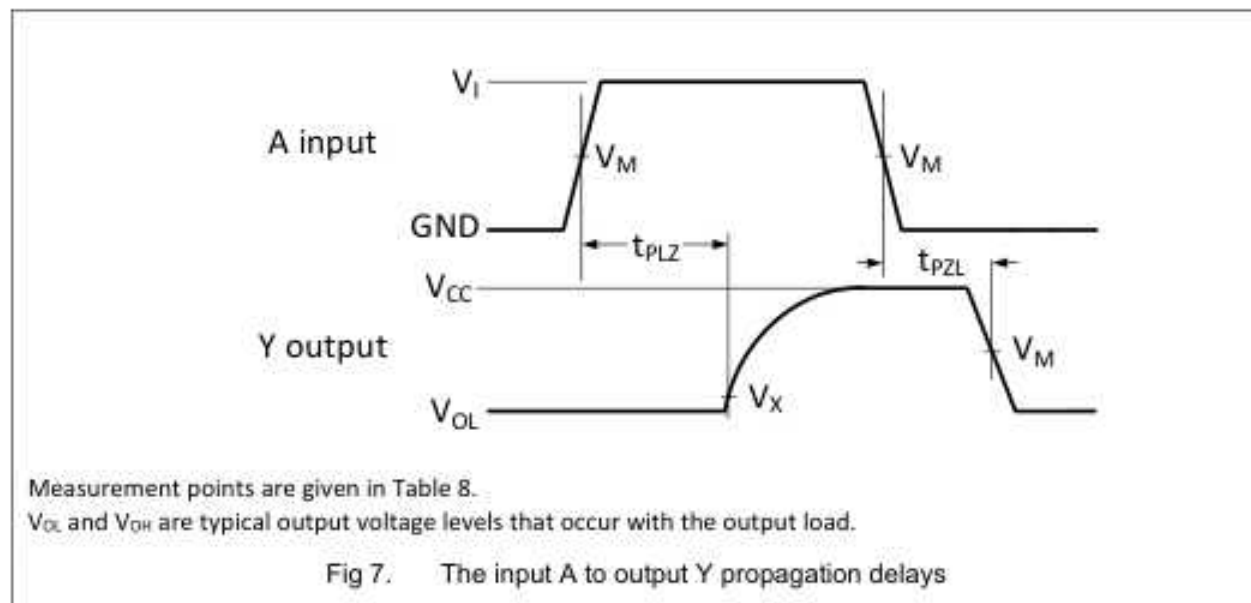
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

$\sum(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

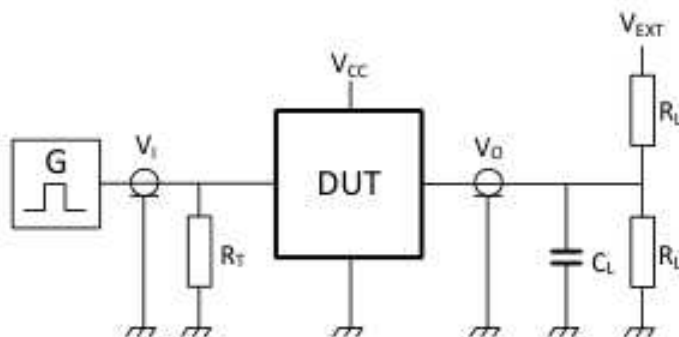
10.1. Waveforms and test circuit


Table 8. Measurement points

Supply voltage	Input		Output	
V_{CC}	V_M	V_I	V_M	V_X
0.8 V to 1.6 V	$0.5V_{CC}$	V_{CC}	$0.5V_{CC}$	$V_{OL} + 0.1 \text{ V}$
1.65 V to 2.7 V	$0.5V_{CC}$	V_{CC}	$0.5V_{CC}$	$V_{OL} + 0.15 \text{ V}$
3.0 V to 3.6 V	$0.5V_{CC}$	V_{CC}	$0.5V_{CC}$	$V_{OL} + 0.3 \text{ V}$

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Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

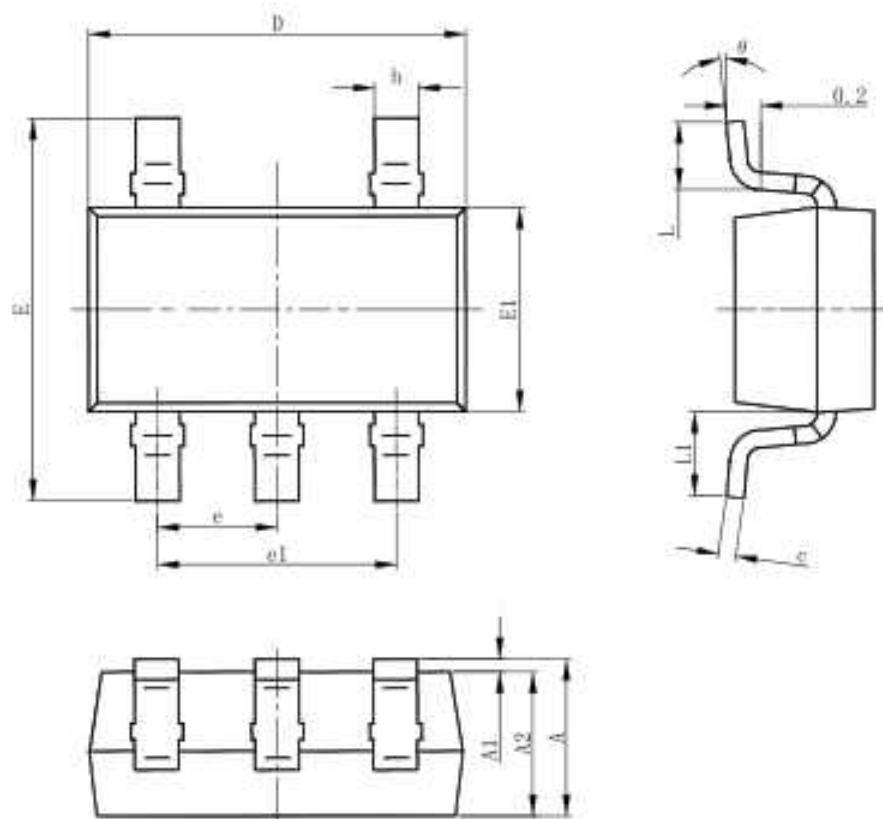
Fig 8. Test circuit for measuring switching times

Table 9. Test data

Supply voltage	Input		Load		VEXT
V_{CC}	V_I	$t_r = t_f$	C_L	R_L	t_{PLZ}, t_{FZL}
0.8 V to 3.6 V	V_{CC}	≤ 2.0 ns	5 pF, 10 pF, 15 pF and 30 pF	5 k Ω	$2V_{CC}$

11. Package Outline

SOT23-5L

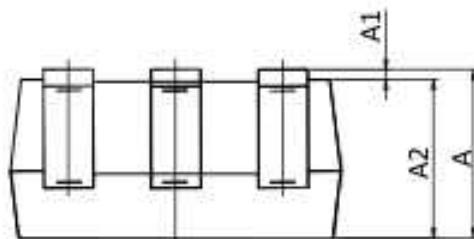
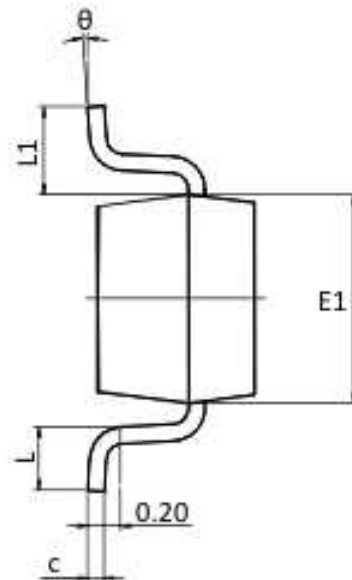
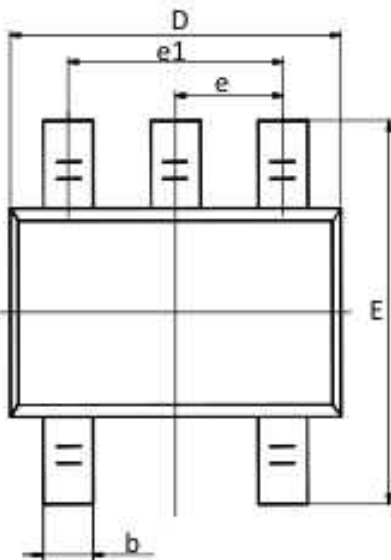


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E1	1.500	1.700	0.059	0.067
E	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
L1	0.600REF.		0.024REF.	
θ	0°	8°	0°	8°

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SOT353

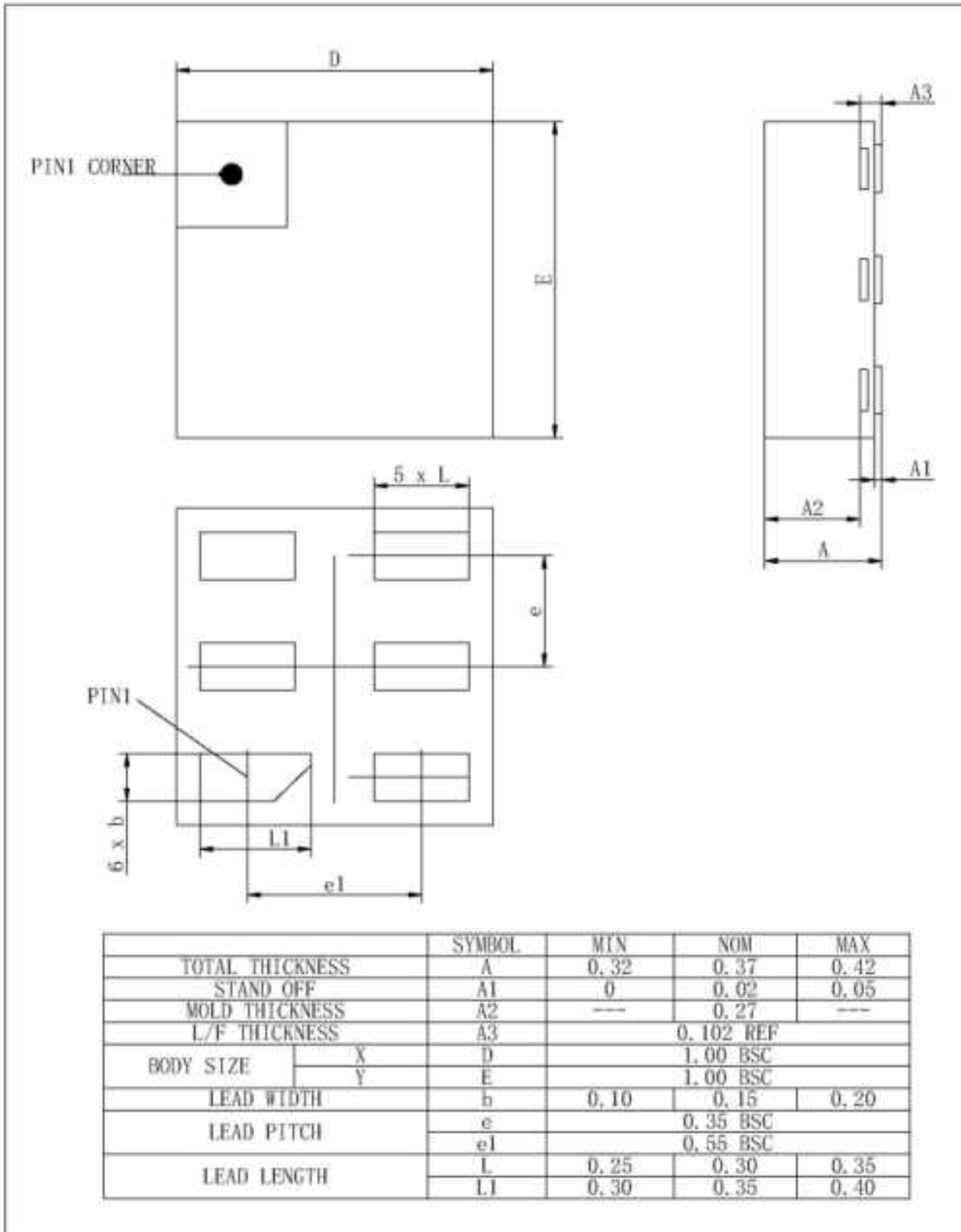


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	0.900	1.100	0.035	0.043
A1	0.000	0.100	0.000	0.004
A2	0.900	1.000	0.035	0.039
b	0.150	0.350	0.006	0.014
c	0.110	0.175	0.004	0.007
D	2.000	2.200	0.079	0.087
E	2.150	2.450	0.085	0.096
E1	1.150	1.350	0.045	0.053
e	0.650 TYP.		0.026 TYP.	
e1	1.200	1.400	0.047	0.055
L	0.260	0.460	0.010	0.018
L1	0.525 REF.		0.021 REF.	
theta	0°	8°	0°	8°

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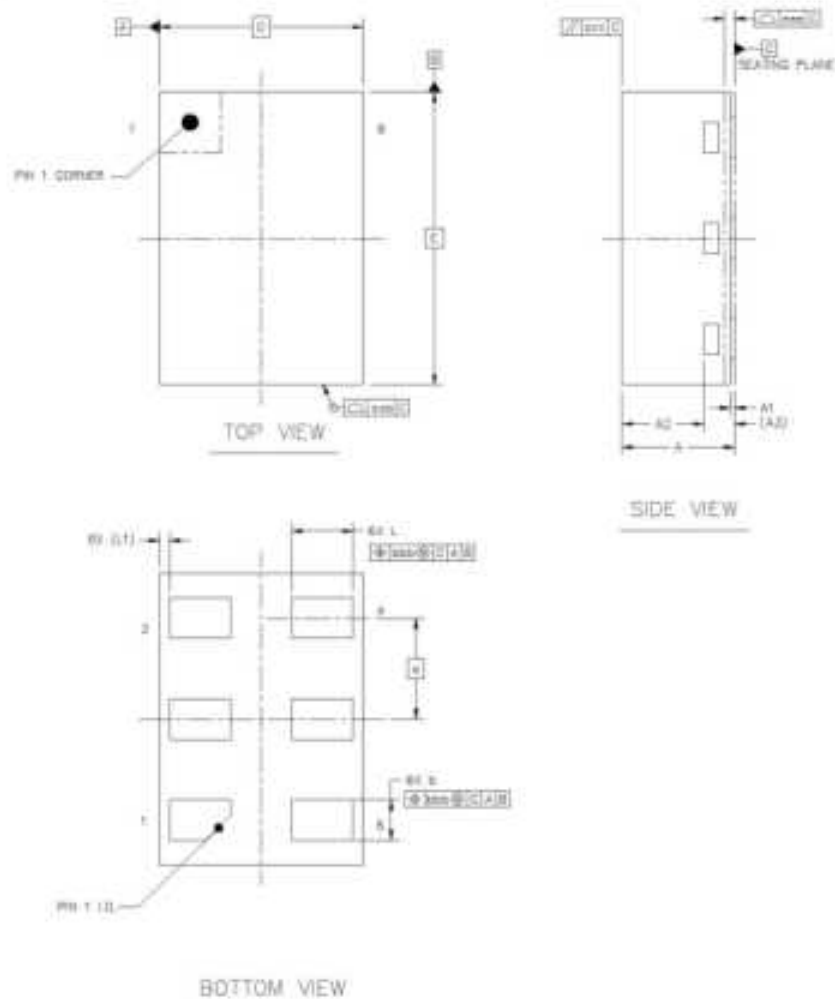
DFN1x1-6L



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Low-power buffer with open-drain output

DFN1x1.45-6L

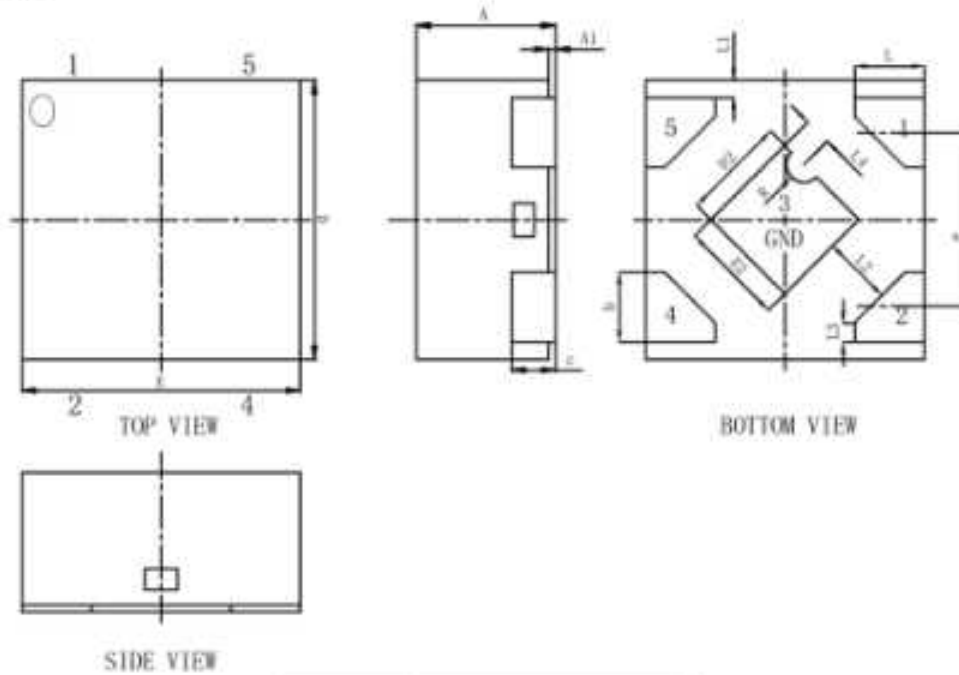


		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.5	0.55	0.6
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.4	---
L/F THICKNESS		A3	0.152 REF		
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D	1 BSC		
	Y	E	1.45 BSC		
LEAD PITCH		e	0.5 BSC		
LEAD LENGTH		L	0.25	0.3	0.35
		L1	0.05 REF		
PACKAGE EDGE TOLERANCE		eed	0.1		
MOLD FLATNESS		ccc	0.1		
COPLANARITY		eee	0.05		
LEAD OFFSET		bbb	0.1		

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DFN0.8x0.8-5L



SYMBOL	MILLIMETER		
	MIN	NOM	MAX
A	0.35	—	0.40
A1	0.00	0.02	0.05
b	0.15	0.20	0.25
c	0.127REF		
D	0.75	0.80	0.85
D2	0.20	0.30	0.40
e	0.50BSC		
E	0.75	0.80	0.85
E2	0.20	0.30	0.40
L	0.15	0.20	0.25
L1	0.05REF		
L2	0.20REF		
L3	0.055REF		
L4	0.10REF		
R	0.05REF		

12. Tape and Reel Information

12.1. Carrier tape dimensions

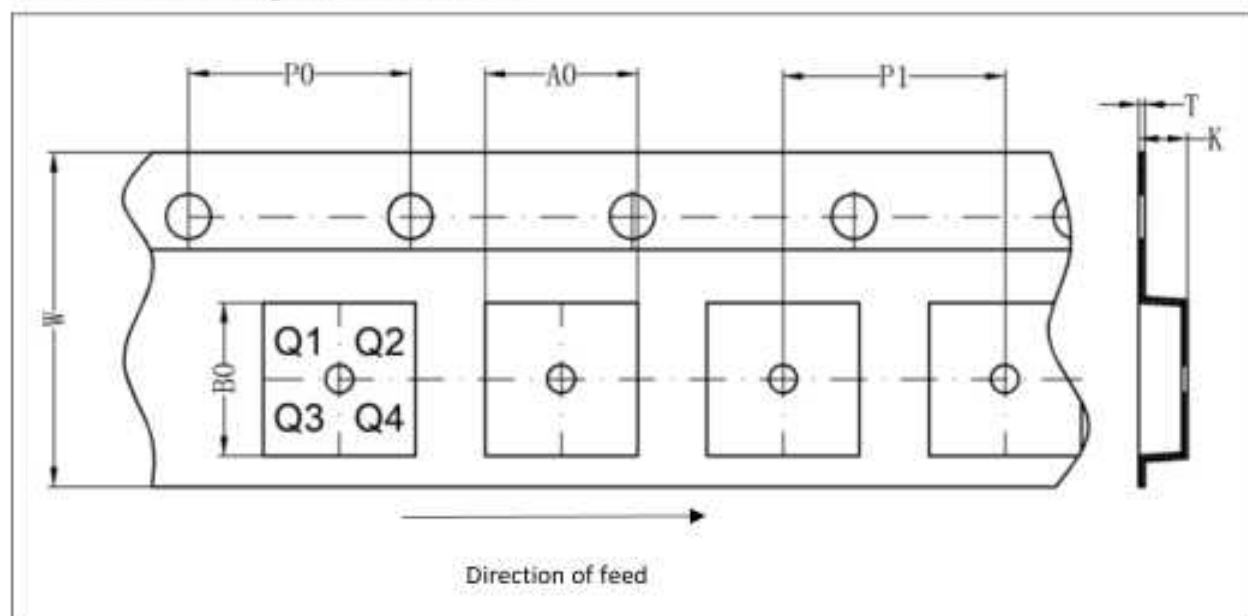


Table 10. Carrier tape dimensions

Type number ending	Package version	A0(mm)	B0(mm)	K(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
GM	DFN1x1.45-6L	1.15	1.6	0.75	0.2	4	8	4	Q1
GS	DFN1x1-6L	1.16	1.16	0.5	0.23	4	8	4	Q1
GW	SOT353	2.55	2.55	1.2	0.2	4	8	4	Q3
GV	SOT23-5L	3.23	3.17	1.37	0.25	4	8	4	Q3
GX	DFN0.8x0.8-5L	0.91	0.91	0.5	0.2	2	8	4	Q1

12.2. Reel and box dimensions

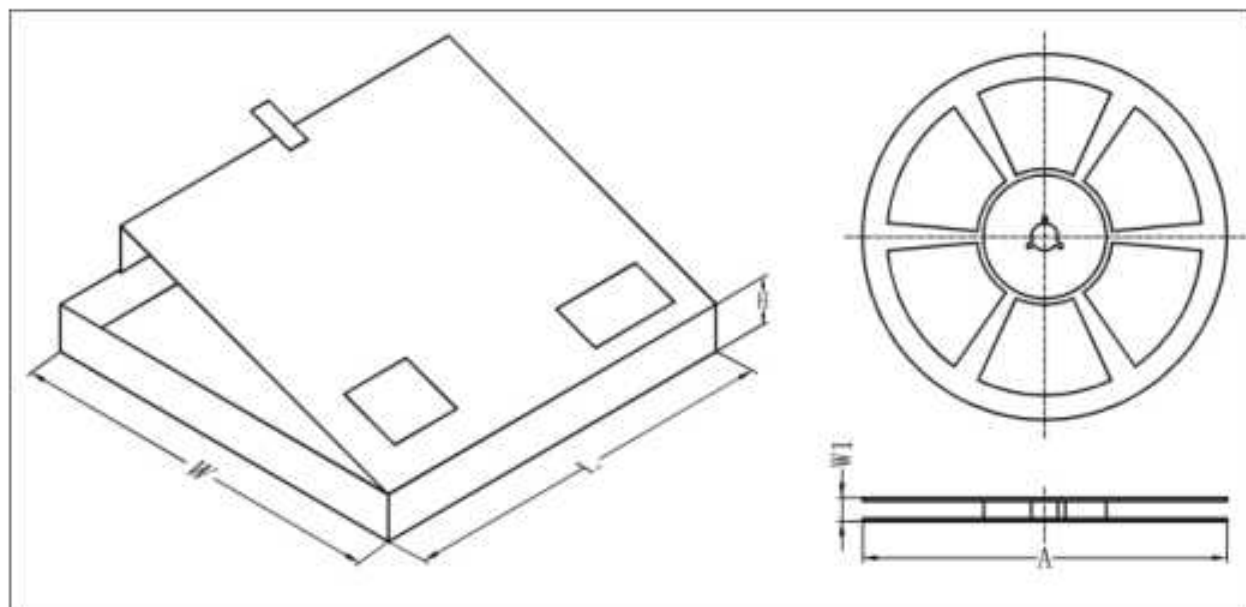


Table 11. Dimensions and quantities

Type number ending	Package version	MSL	Reel Dimension A (mm)	Max Reel Width W1 (mm)	SPQ (pcs)[1]	Reels per box	Outer box dimensions L×W×H(mm)[2]
GM	DFN1x1.45-6L	MSL1	180	13.2	3000	1	210x200x40
GS	DFN1x1-6L	MSL1	180	13.2	3000	1	210x200x40
GW	SOT353	MSL1	180	13.2	3000	1	210x200x40
GV	SOT23-5L	MSL1	180	13.2	3000	1	210x200x40
GX	DFN0.8x0.8-5L	MSL1	180	13.2	3000	1	210x200x40

[1] Packing quantity dependent on specific product type. Please contact your local Energymath representative for ordering.

[2] Dimensions for reference only.

13. Abbreviations

Table 12. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
MM	Machine Model

14. Revision History

Table 13. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74AUP1G07 Rev0.1	Jun 16, 2026	Draft datasheet		